Homework 3

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Set up

```r
# read in tidyverse for data wrangling and visualization
library(tidyverse)
# read in here for reproducible file paths
library(here)
# read in janitor for cleaning column names
library(janitor)

# read in kelp data from data folder
kelp <- read_csv(here("data", "temp-kelp.csv"))

# read in personal data from data folder
my_data <- read_csv(here("data", "MyData_Homework3.csv"))
```

Problem 1: Giant kelp fronds

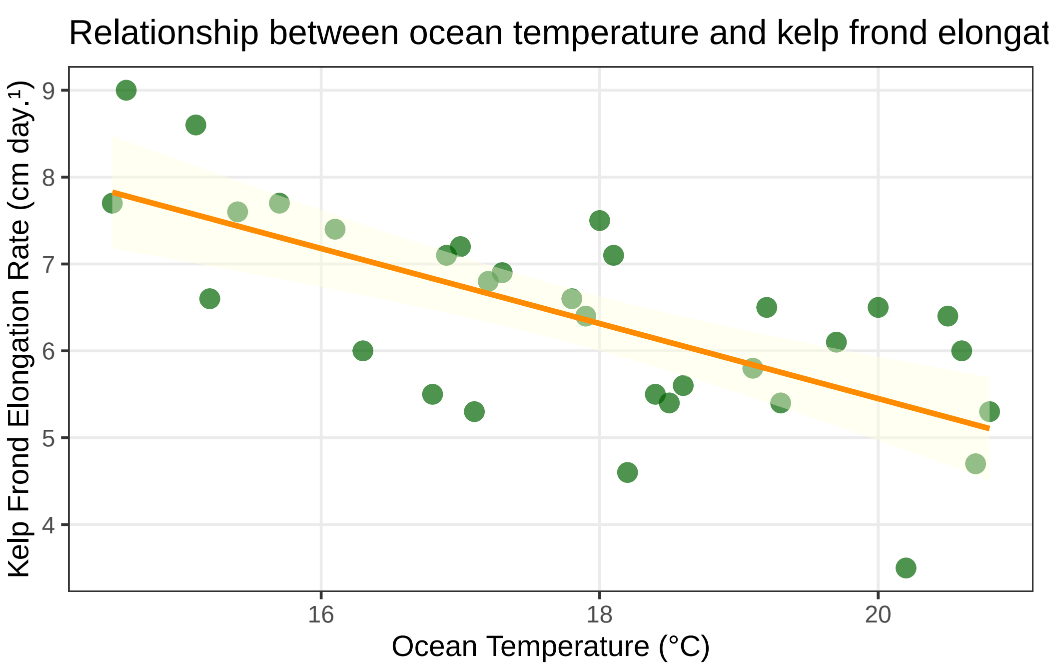
1a. An appropriate test

The two appropriate tests to determine the strength of the relationship between ocean temperature and giant kelp frond elongation rate are the Pearson correlation coefficient and the Spearman rank correlation coefficient. The Pearson correlation measures the strength and direction of a linear relationship between two continuous variables and assumes that both variables are normally distributed. The Spearman rank correlation is a non-parametric alternative that measures the strength of a monotonic relationship between ocean temperature and elongation rate without assuming normality, making it appropriate if the normality assumption for Pearson is violated.

1b. Create a visualization

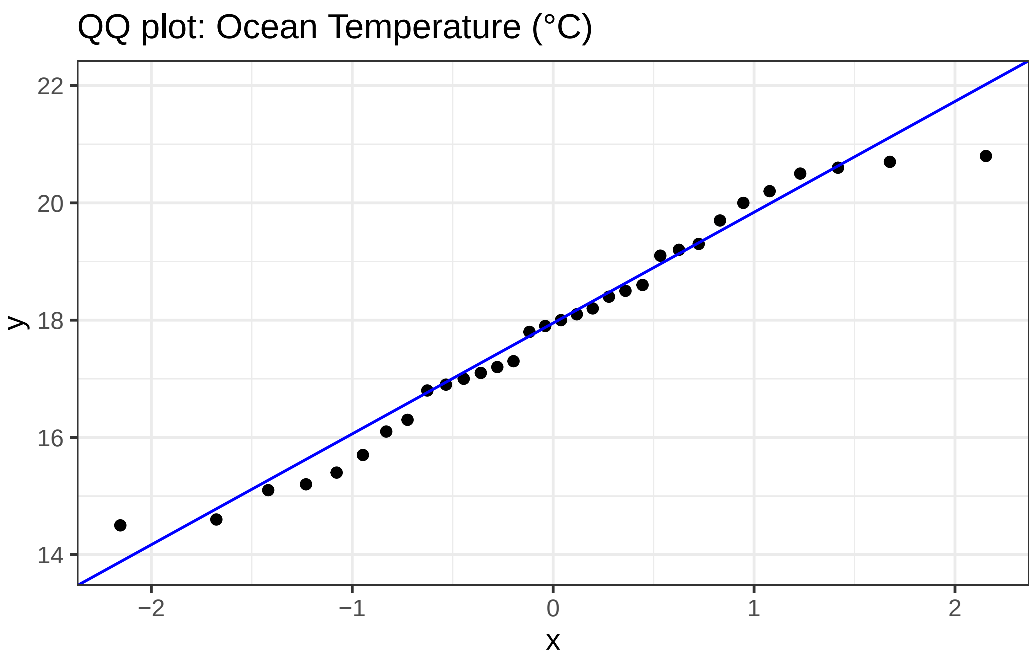
```r
# create scatterplot of ocean temperature vs kelp frond elongation rate
ggplot(data = kelp, # use kelp data frame
       aes(x = temp_c, # ocean temperature on x-axis
           y = kelp_elong)) + # elongation rate on y-axis
  geom_point(color = "darkgreen", # non-default point color
             size = 3,
             alpha = 0.7) + # slight transparency on points
  geom_smooth(method = "lm", # add linear trend line
              color = "darkorange", # non-default line color
              fill = "lightyellow", # confidence interval fill color
              se = TRUE) + # show confidence interval around line
  labs(x = "Ocean Temperature (°C)", # label x-axis with units
       y = "Kelp Frond Elongation Rate (cm day ¹)", # label y-axis with units
       title = "Relationship between ocean temperature and kelp frond elongation rate") +
  theme_bw() + # non-default theme
  theme(panel.grid.minor = element_blank()) # remove minor gridlines to reduce clutter
```

```
`geom_smooth()` using formula = 'y ~ x'
```

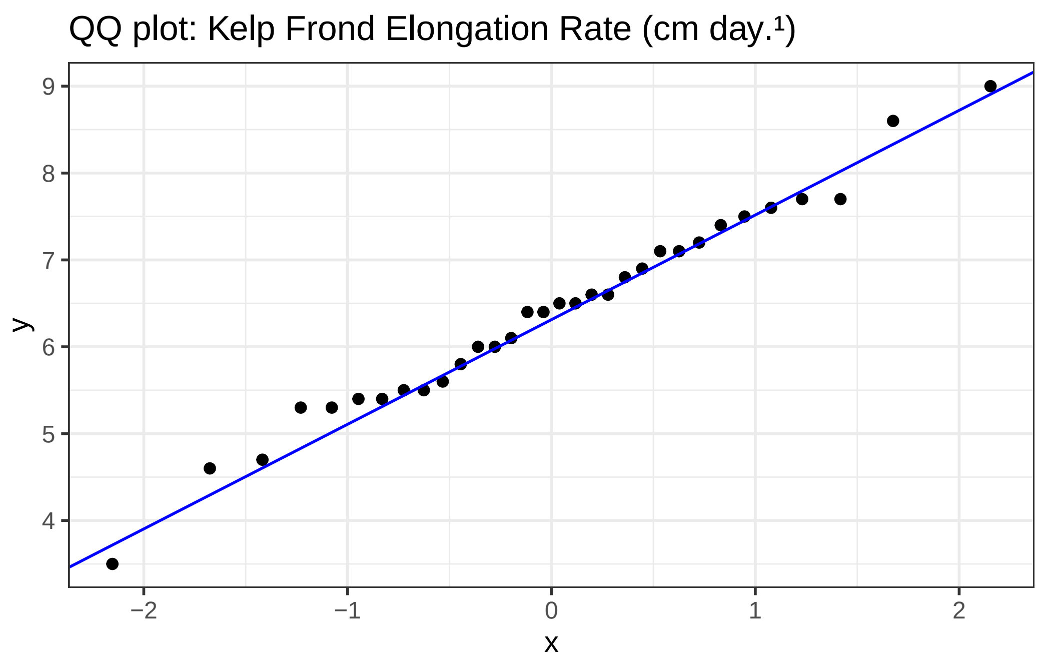


1c.Check your assumptions and run your test

```r
# check normality of ocean temperature using a QQ plot
ggplot(data = kelp,
       aes(sample = temp_c)) + # check temperature variable
  geom_qq() + # plot sample quantiles against theoretical quantiles
  geom_qq_line(color = "blue") + # add reference line for normality
  labs(title = "QQ plot: Ocean Temperature (°C)") +
  theme_bw() # non-default theme
```




```r
# check normality of kelp elongation rate using a QQ plot
ggplot(data = kelp,
       aes(sample = kelp_elong)) + # check elongation rate variable
  geom_qq() + # plot sample quantiles against theoretical quantiles
  geom_qq_line(color = "blue") + # add reference line for normality
  labs(title = "QQ plot: Kelp Frond Elongation Rate (cm day ¹)") +
  theme_bw() # non-default theme
```




```
# run Pearson correlation test between ocean temperature and kelp elongation rate
cor.test(kelp$temp_c, # predictor variable: ocean temperature
         kelp$kelp_elong, # response variable: kelp frond elongation rate
         method = "pearson") # specify Pearson method for linear relationship
```

```
    Pearson's product-moment correlation

data:  kelp$temp_c and kelp$kelp_elong
t = -5.189, df = 30, p-value = 1.366e-05
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.8359665 -0.4460157
sample estimates:
       cor
-0.6877489
```

1d.

To evaluate the strength of the relationship between ocean temperature and giant kelp frond elongation rate, I used a Pearson correlation coefficient because both variables are continuous and approximately normally distributed, and the relationship appears linear. The results indicate a significant negative relationship between ocean temperature and kelp frond elongation rate (Pearson's $r = -0.68$, $t(30) = -5.19$, $p < 0.001$).